

Yeqiao Fu

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EDUCATION

University of Hong Kong (HKU)

Hong Kong, China

BEng in Computer Science, minor in Finance | CGPA: 3.64/4.30, Ranking: Top 10%

Sept. 2022- Jun. 2026 (Expected)

University of California, Davis (UCD)

Davis, CA

Exchange, major in Computer Science

Jan. 2025- Jun. 2025

- **Relevant Coursework:** Artificial Intelligence, Linux-Based Data Analysis, R-Based Data Visualization and Analysis, Computer Programming in Python, Computer Programming in C++, Object-Oriented Programming and Java, Data Structures and Algorithms, Probability and Statistics, Linear Algebra, Calculus and Ordinary Differential Equations

PUBLICATION

R. Cao, F. Lei, H. Wu, J. Chen, Y. Fu, et al. Spider2-V: How Far Are Multimodal Agents from Automating Data Science and Engineering Workflows, NeurIPS, 2024.

Research Experience

Table Reasoning Evaluation Framework

Remote

Research Assistant, supervised by Prof. Philipp Koehn at Johns Hopkins University

Jun. 2025 – Oct. 2025

- Designed and built TABULA-R², a benchmark of 129 real-world tables (from Our World in Data) with 252 manually authored and verified QA pairs across 8 domains, each annotated for answerability, complexity, reasoning type, and domain.
- Implemented a unified PLAN/END protocol and a lightweight domain-specific language (DSL) for executable table reasoning, together with a validator-executor that converts LLM outputs into $\approx 3,000$ executable DSL steps with per-step logging and structured error taxonomy.
- Developed a fully reproducible experimental pipeline (YAML configurations, schema validation, randomized batching, visualization) supporting single-table, multi-table, and distractor-augmented settings under a common interface.
- Conducted controlled comparison experiments over five local LLM families (Llama3, Llama3.2, Gemma, Qwen, CodeLlama) and multiple prompt settings (few-shot $k=0/1/3$, Chain-of-Thought inclusion, observation modes: header_1, header_5, full_table), revealing strong protocol-alignment effects (zero-shot + header_1 outperforming few-shot/CoT).
- Led the project end-to-end — from dataset curation and toolkit engineering to analysis and visualization — culminating in an 86-page technical report commended by my advisor described as “publishable” and “at the level of a senior graduate student.”
- Open-sourced the entire codebase, benchmark, and experiment logs to support reproducible evaluation of table reasoning: <https://github.com/fyq5166/TABULA-R2>

Interactive Website Modeling and Trajectory Generation

XLANG Lab HKU

Research assistant, supervised by Prof. Tao Yu at HKU

Jul. 2024 - Nov. 2024

- Constructed interactive world models for websites to enable AI agents to explore and record interaction outcomes, capturing structured intra- and inter-page data with ally-trees and URL maps.
- Designed a comprehensive pipeline for trajectory generation, including exploration, task execution, and evaluation.
- Reduced redundancy in web interaction via component classification (mechanical & semantic algorithms) and URL navigation reduction.
- Achieved 100% grounding and replay success using ally-tree-based localization, collecting high-level descriptions and step-by-step trajectories to guide task generation.
- Designed an AI evaluator to assess trajectories based on Completeness, Complexity, Conciseness, Concreteness, and Diversity, achieving alignment with human evaluation (68–74%) using 4omini.
- Developed modular and extensible project structures with automated pipelines from configuration to evaluation.

- Exploring vision-based approaches for web interaction and evaluating world model methods on established benchmarks to refine agent performance.

Spider2-V: How Far Are Multi-modal Agents from Automating Data Science and Engineering Workflows XLANG Lab HKU

Research assistant, supervised by Prof. Tao Yu at HKU

Mar. 2024 - Jun. 2024

- Designed and implemented a benchmark to evaluate multi-modal agents' ability to automate data science workflows with 494 real-world tasks spanning 20 enterprise-level applications.
- Trained and evaluated vision-language models in Linux-based environments for joint code-generation and GUI-operations tasks.
- Developed novel grounding algorithms to enhance performance on multi-step data science tasks, addressing challenges in executing fine-grained, knowledge-intensive GUI interactions.
- Applied advanced prompt-tuning techniques to refine model responses, improving accuracy and context awareness from 14.0% to 35.0% in professional tool interactions.
- Co-authored Spider2-V: How Far Are Multimodal Agents from Automating Data Science and Engineering Workflows, featured as a spotlight presentation at the NeurIPS D&B track in September 2024.

Machine Learning and the Application within Cyber Incident Response

OSRP Cambridge

Independent Research, supervised by Prof. Kieren Lovell at the University of Cambridge

Jun. 2023 - Sep. 2023

- Developed robust text preprocessing workflows using the NLTK library, including tokenization, POS tagging, lemmatization, and stop-word filtering, while optimizing downstream text vectorization for improved model performance.
- Implemented state-of-the-art GloVe and Word2Vec models to create comprehensive text representations, capturing intricate semantic relationships and contextual nuances in textual data.
- Designed a six-class scam-evaluation system based on the STP-II model, enabling detailed analysis and categorization of potential threats.
- Utilized Sklearn's multi-output regression and iterative model tuning to forecast scam threat levels, achieving 90% accuracy on low-abstraction scam attributes.

INTERNSHIP

ByteDance · Volcano Engine Ark Group

Beijing

AI Agent Engineering Intern at Data Dept

Jul. 2025 – Aug. 2025

- Designed dual-track Arkitect Cookbooks (human-oriented Jupyter Notebook paired with agent-oriented Python script) to demonstrate end-to-end workflows; open-source PR accepted into the 1.7 k-star Arkitect repository.
- Built a modular evaluation framework for agent tool-calling, enabling dynamic tool registration (local, API, MCP) configurable test suite design, and hybrid validation (rule-based + LLM-as-judge) with structured logs for prompt-engineering analysis.
- Automated incident support by (1) analyzing 1,000+ records to link incidents with Grafana dashboards, (2) implementing secure image-based context management via TOS buckets, and (3) constructing vector knowledge base.
- Improved automated response effectiveness via prompt engineering (PE) and refined trigger logic with layered strategies to reduce repetitive Q&A and accelerated incident resolution.

Skillset & Interests

Languages: Mandarin (Native), Cantonese (Intermediate), English (fluent, IELTS: 8, TOEFL: 110, GRE: 326)

Technical Skills: Python (primary), C/C++, Java, R; SQL (MySQL, Snowflake); Linux, Docker, Git, Airflow, Grafana, vector search; LLM / agent tooling (PLAN/END protocols, tool-calling, LLM-as-judge); HTML/DOM & GUI automation; Matplotlib, PlantUML, LaTeX.